

Magnetic Field and Magnetic Forces

Forces on moving charges: circular orbits, mass spectrometers, cyclotrons, motors, and the physics inside an MRI scanner.

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Magnetism was known for thousands of years through lodestones and compass needles, but its true nature emerged only in the nineteenth century: **magnetic forces act on moving charges**, and magnetic fields are produced by moving charges. This chapter defines the magnetic field through the force it exerts, the Lorentz force. It then follows charged particles round circular and helical paths, and uses this to explain velocity selectors, mass spectrometers and cyclotrons. It next finds the forces and torques on current-carrying wires and coils, which drive every electric motor and loudspeaker. The physics here also underlies some of medicine's most powerful tools: MRI, proton therapy and Doppler-free electromagnetic flowmeters.

YOU WILL BE ABLE TO

- Define the magnetic field $\vec{B}$ through the force $\vec{F} = q\vec{v} \times \vec{B}$, and use the right-hand rule.
- Explain why magnetic forces do no work, and analyse circular and helical motion of charged particles.
- Analyse velocity selectors, mass spectrometers, cyclotrons and the Hall effect.
- Calculate forces on current-carrying conductors, $\vec{F} = I\vec{L} \times \vec{B}$.
- Calculate the torque and energy of a current loop in a field, using the magnetic dipole moment, and apply them to motors and meters.
- Describe the magnetic physics behind MRI and electromagnetic blood-flow measurement.

THINK FIRST

In an MRI scanner the magnetic field is about 3 T, some 60 000 times stronger than the Earth's. A patient lies perfectly still inside it and feels nothing. Yet a steel oxygen cylinder brought into the room can be pulled in violently enough to kill. Why does the field affect the cylinder so dramatically but not the patient's body, which is full of moving charged ions?

Two different effects are involved. The force on a **moving charge**, $q\vec{v} \times \vec{B}$, is tiny for ions drifting slowly in the body, and it does no work, so the patient feels nothing. Steel, however, is **ferromagnetic**: the field lines up its countless atomic magnetic moments, turning the cylinder into a strong magnet, which is then pulled hard towards the region where the field is strongest. Human tissue is only very weakly magnetic. That is why MRI safety screening is all about ferromagnetic objects (implants, tools, cylinders), not about the field acting on the body itself.

§18.1

The magnetic field

A **magnetic field** $\vec{B}$ exists wherever a moving charge feels a force that depends on its velocity. Bar magnets have north and south poles. Like poles repel and unlike poles attract, and a compass needle, itself a small magnet, points along the field. Magnetic field lines leave north poles and enter south poles outside a magnet, and continue *through* the magnet to form **closed loops**. Unlike electric field lines, they never begin or end: no isolated magnetic “charge” (monopole) has ever been found. This is expressed by **Gauss's law for magnetism**:

$$\oint \vec{B} \cdot d\vec{A} = 0.$$

The magnetic flux through any closed surface is zero.

The SI unit of $\vec{B}$ is the **tesla**: $1 \text{ T} = 1 \text{ N}/(\text{A m})$. The older gauss is still common, with $1 \text{ G} = 10^{-4} \text{ T}$.

SOURCE	FIELD
Brain activity (measured by MEG)	10^{-13} T
Earth's surface	$3\text{--}6 \times 10^{-5} \text{ T}$ (0.3–0.6 G)

SOURCE	FIELD
Fridge magnet	about 0.01 T
Neodymium magnet surface	about 1 T
Clinical MRI	1.5–3 T (research systems: 7–11.7 T)
Strongest laboratory steady fields	about 45 T
Neutron star surfaces	10^8 – 10^{11} T

§18.2

The magnetic force on a moving charge

A charge q moving with velocity $\vec{v}$ through a magnetic field $\vec{B}$ feels the **magnetic force**

$$\vec{F} = q\vec{v} \times \vec{B}, \quad F = |q|vB \sin \theta,$$

where θ is the angle between $\vec{v}$ and $\vec{B}$. With an electric field also present, the total is the **Lorentz force**:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}).$$

Direction (right-hand rule). Point your fingers along $\vec{v}$ and curl them towards $\vec{B}$. Your thumb gives the direction of $\vec{v} \times \vec{B}$. Reverse it for a negative charge.

KEY IDEA

Key features of the magnetic force.

1. It is zero for a charge at rest, and for a charge moving parallel to $\vec{B}$.
2. It is always **perpendicular to the velocity**, so it **does no work** and cannot change a particle's speed or kinetic energy. It only bends the path.
3. It is perpendicular to $\vec{B}$ as well, which is unlike the electric force.

WORKED EXAMPLE 18.1

Magnetic forces on particles

(a) A proton moves at 3.0×10^6 m/s perpendicular to a 0.50 T field. Find the force on it and its acceleration. (b) An electron moves at 1.0×10^6 m/s perpendicular to the Earth's field, 5.0×10^{-5} T. Find the force.

Solution

(a) For the proton:

$$F = evB = (1.602 \times 10^{-19})(3.0 \times 10^6)(0.50) = 2.4 \times 10^{-13} \text{ N}, \quad a = \frac{F}{m_p} = 1.4 \times 10^{14} \text{ m/s}^2$$

(b) For the electron:

$$F = (1.602 \times 10^{-19})(1.0 \times 10^6)(5.0 \times 10^{-5}) = 8.0 \times 10^{-18} \text{ N}.$$

This is tiny, but it produces an acceleration of about 10^{13} m/s^2 , enough to bend electron beams noticeably. Old CRT televisions had to be adjusted for the local direction of the Earth's field.

§18.3

Motion of charged particles in a uniform field

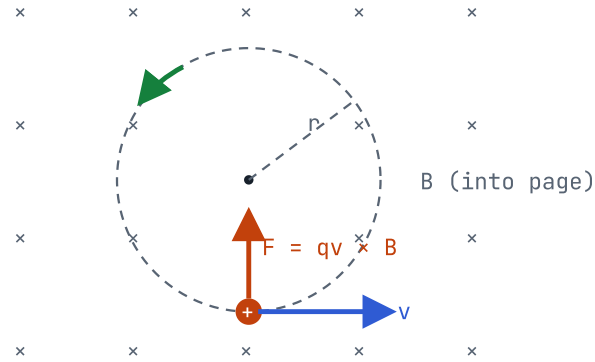
Velocity perpendicular to $\vec{B}$. The force has constant magnitude $|q|vB$ and is always perpendicular to $\vec{v}$, so the particle moves in a **circle** at constant speed. The magnetic force supplies the centripetal force:

$$|q|vB = \frac{mv^2}{r} \implies r = \frac{mv}{|q|B} = \frac{p}{|q|B}.$$

The angular frequency and period of the motion are

$$\omega_c = \frac{|q|B}{m}, \quad T = \frac{2\pi m}{|q|B}.$$

ω_c is the **cyclotron frequency**. It is **independent of the speed** (as long as $v \ll c$): faster particles move on proportionally larger circles and take the same time to go round.



In a uniform field pointing into the page, a positive charge circles anticlockwise. The magnetic force always points towards the centre, at right angles to the velocity.

Velocity at an angle to $\vec{B}$. Split the velocity into components. The component perpendicular to $\vec{B}$ produces circular motion of radius $mv_{\perp}/|q|B$. The component parallel to $\vec{B}$ is unaffected. The path is a **helix** whose pitch (the distance advanced per turn) is $v_{\parallel}T$. Charged particles from the Sun spiral along the Earth's field lines in this way towards the poles, where they strike the upper atmosphere and produce the **aurora**.

WORKED EXAMPLE 18.2

An electron circling in a field

An electron moves at 2.0×10^6 m/s perpendicular to a 1.0 mT field. Find the radius of its orbit and its cyclotron frequency.

Solution

$$r = \frac{m_e v}{eB} = \frac{(9.11 \times 10^{-31})(2.0 \times 10^6)}{(1.602 \times 10^{-19})(1.0 \times 10^{-3})} = 1.1 \times 10^{-2} \text{ m} = 1.1 \text{ cm.}$$

$$f = \frac{eB}{2\pi m_e} = \frac{(1.602 \times 10^{-19})(1.0 \times 10^{-3})}{2\pi(9.11 \times 10^{-31})} = 2.8 \times 10^7 \text{ Hz} = 28 \text{ MHz.}$$

The electron cyclotron frequency is 28 GHz per tesla. Microwave ovens and some fusion-plasma heating systems exploit this resonance.

§18.4

Applications of charged-particle motion

The velocity selector

Crossed electric and magnetic fields, perpendicular to each other and to the beam, push in opposite directions on a charge. Only particles whose electric and magnetic forces balance, $qE = qvB$, pass straight through:

$$v = \frac{E}{B}.$$

The selected speed does not depend on the particle's charge or mass.

The mass spectrometer

Ions of charge q are accelerated through a potential difference V (gaining $qV = \frac{1}{2}mv^2$), then enter a uniform field B and bend on a semicircle of radius

$$r = \frac{mv}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}}.$$

Ions of different mass land at different places, so the instrument sorts them by their mass-to-charge ratio.

WORKED EXAMPLE 18.3

Separating carbon-12 and carbon-14

Singly charged carbon ions are accelerated through 1000 V and bent by a 0.10 T field. Find the radii of the ^{12}C and ^{14}C paths, and their separation after a semicircle. ($1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$.)

Solution

For ^{12}C , $m = 12(1.66 \times 10^{-27}) = 1.99 \times 10^{-26} \text{ kg}$:

$$r_{12} = \frac{1}{0.10} \sqrt{\frac{2(1.99 \times 10^{-26})(1000)}{1.602 \times 10^{-19}}} = \frac{1}{0.10} \sqrt{2.49 \times 10^{-4}} = 0.158 \text{ m}.$$

Since $r \propto \sqrt{m}$:

$$r_{14} = 0.158 \sqrt{14/12} = 0.170 \text{ m}.$$

After a semicircle, the ions land a diameter away from the entrance, so they are separated by $2(r_{14} - r_{12}) = 2.5 \text{ cm}$.

Evaluate. *Accelerator mass spectrometry* counts individual ^{14}C atoms in this way, even though they make up only about one part in 10^{12} of the carbon. It gives radiocarbon dates from milligram samples. In medicine and pharmacology, mass spectrometers identify drugs and metabolites, screen newborns for metabolic diseases, and identify bacteria in minutes.

The cyclotron

A cyclotron accelerates charged particles using two hollow D-shaped electrodes (“dees”) inside a magnetic field. The particles circle inside the dees, and an alternating voltage across the gap between them gives a kick each time they cross it. Because the orbital period does not depend on speed, a *fixed* frequency $f = qB/2\pi m$ stays in step with the particles as they spiral outward. They leave at the outer radius R with

$$K_{\max} = \frac{q^2 B^2 R^2}{2m}.$$

WORKED EXAMPLE 18.4

A cyclotron for proton therapy

A cyclotron has a 2.0 T field and an extraction radius of 0.50 m. Find the radio frequency it must use for protons, and the maximum proton energy in MeV.

Solution

$$f = \frac{eB}{2\pi m_p} = \frac{(1.602 \times 10^{-19})(2.0)}{2\pi(1.673 \times 10^{-27})} = 3.0 \times 10^7 \text{ Hz} = 30 \text{ MHz}.$$

$$K = \frac{e^2 B^2 R^2}{2m_p} = \frac{(1.602 \times 10^{-19})^2 (4.0)(0.25)}{2(1.673 \times 10^{-27})} = 7.7 \times 10^{-12} \text{ J} = 48 \text{ MeV}.$$

Evaluate. Proton therapy needs 70–250 MeV to reach deep tumours. Clinical machines therefore use superconducting cyclotrons with larger fields, or synchrotrons. At these energies the protons are mildly relativistic, so the field is shaped to keep them in step. Proton beams deposit most of their energy at a well-defined depth (the *Bragg peak*), which spares healthy tissue beyond the tumour. This is especially valuable for children and for tumours near the spinal cord or eyes.

The Hall effect

When a current flows along a conducting strip in a perpendicular magnetic field, the carriers are pushed to one side of the strip. Charge builds up until the resulting sideways electric field balances the magnetic force. The voltage across the strip is the **Hall voltage**:

$$V_H = \frac{IB}{nqt},$$

where t is the thickness of the strip in the direction of $\vec{B}$. Its sign reveals whether the carriers are positive or negative, and its size gives the carrier density n . Hall sensors in semiconductors, where n is small and so V_H is large, measure magnetic fields, detect wheel rotation for ABS brakes, and act as the electronic compass in phones.

§18.5

Force on a current-carrying conductor

A wire carrying current I contains moving charges, so it feels a force in a magnetic field. For a straight segment of length L in a uniform field:

$$\vec{F} = I\vec{L} \times \vec{B}, \quad F = ILB \sin \theta,$$

where $\vec{L}$ points along the current. For a curved wire, add up the forces on each small segment: $\vec{F} = \int I d\vec{l} \times \vec{B}$.

Derivation. A segment of length L and cross-section A contains nAL carriers, each feeling $q\vec{v}_d \times \vec{B}$. The total force is $nALq\vec{v}_d \times \vec{B} = I\vec{L} \times \vec{B}$, using $I = nqv_dA$.

Useful fact: in a *uniform* field, the net force on any **closed** loop of current is zero. A curved wire between two points feels the same force as a straight wire between those points (Problem P18.15).

WORKED EXAMPLE 18.5

Levitating a rod

A horizontal copper rod of mass 50 g and length 20 cm lies across two rails in a horizontal magnetic field of 0.50 T, perpendicular to the rod. What current is needed to make the rod float?

Solution

The magnetic force must balance the rod's weight:

$$ILB = mg \Rightarrow I = \frac{mg}{LB} = \frac{(0.050)(9.8)}{(0.20)(0.50)} = 4.9 \text{ A.}$$

The current direction must be chosen so that $\vec{L} \times \vec{B}$ points up.

§18.6

Torque on a current loop

A rectangular loop carrying current I in a uniform field feels zero net force, but it does feel a **torque**. Forces on opposite sides are equal and opposite but act along different lines, forming a couple. For a flat coil of N turns enclosing area A , define the **magnetic dipole moment**

$$\vec{\mu} = NI\vec{A},$$

a vector perpendicular to the plane of the coil. Its direction is given by the right-hand rule: curl your fingers along the current, and your thumb points along $\vec{\mu}$. Then

$$\vec{\tau} = \vec{\mu} \times \vec{B}, \quad \tau = \mu B \sin \theta, \quad U = -\vec{\mu} \cdot \vec{B}.$$

These have exactly the same form as for an electric dipole in an electric field (Chapter 13). The torque tries to line $\vec{\mu}$ up with $\vec{B}$.

WORKED EXAMPLE 18.6

A motor coil

A motor coil has 100 turns, each enclosing 25 cm^2 , and carries 2.0 A in a 0.50 T field. Find its magnetic moment and the maximum torque.

Solution

$$\mu = NIA = 100(2.0)(25 \times 10^{-4}) = 0.50 \text{ A m}^2, \quad \tau_{\max} = \mu B = 0.50(0.50) = 0.25 \text{ N m.}$$

Evaluate. In a **DC motor**, a commutator reverses the current in the coil every half-turn, just as $\vec{\mu}$ swings past $\vec{B}$. The torque therefore always acts in the same rotational direction, and the motor keeps spinning. **Loudspeakers** use the force on a coil of wire in the field of a permanent magnet to drive a cone back and forth. Moving-coil **galvanometers** balance

$\tau = NIAB$ against the twist of a spring, so the needle's deflection is proportional to the current.

IN PRACTICE

Magnetism in medicine.

- **MRI.** Hydrogen nuclei (protons) are tiny magnetic dipoles. In the scanner's strong field B_0 , they precess about the field at the **Larmor frequency**, $f = \gamma B_0 / 2\pi$, which is **42.58 MHz per tesla** for protons (64 MHz at 1.5 T). A radio pulse at this frequency tips them over, and as they relax they emit radio signals. Gradient coils make B_0 vary slightly with position, so that each location in the body “sings” at its own frequency. That is how the image is built up (Problem P18.12). Different tissues relax at different rates, which gives MRI its superb soft-tissue contrast, all without ionising radiation.
- **Electromagnetic flowmeters.** Blood is a conducting fluid. When it flows through a field, the $q\vec{v} \times \vec{B}$ force pushes positive and negative ions to opposite sides of the vessel, producing a voltage $V = Bdv$ across a vessel of diameter d . Measuring this voltage gives the flow speed directly (Problem P18.11). The same principle is used for industrial flow measurement of water and slurries.
- **Magnetoencephalography (MEG)** detects the femtotesla fields produced by currents in the brain, using superconducting or optically pumped sensors.

CHAPTER SUMMARY

- Magnetic force: $\vec{F} = q\vec{v} \times \vec{B}$, with magnitude $|q|vB \sin \theta$. Lorentz force: $q(\vec{E} + \vec{v} \times \vec{B})$. 1 T = 1 N/(A m).
- The magnetic force is perpendicular to $\vec{v}$, so it does no work and the speed is unchanged.
- Circular motion: $r = mv/|q|B$, $\omega_c = |q|B/m$ (independent of speed). At an angle to $\vec{B}$, the path is a helix.
- Velocity selector: $v = E/B$. Mass spectrometer: $r = \sqrt{2mV/q}/B$. Cyclotron: $f = qB/2\pi m$ and $K_{\max} = q^2 B^2 R^2 / 2m$. Hall voltage: $V_H = IB/(nqt)$.
- Force on a wire: $\vec{F} = I\vec{L} \times \vec{B}$. The net force on a closed loop in a uniform field is zero.
- Loop: $\vec{\mu} = NI\vec{A}$, $\vec{\tau} = \vec{\mu} \times \vec{B}$ and $U = -\vec{\mu} \cdot \vec{B}$.
- $\oint \vec{B} \cdot d\vec{A} = 0$: there are no magnetic monopoles.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P18.1

Can a magnetic field change the speed of a charged particle? Can it change the particle's kinetic energy? Its velocity? Explain.

P18.2

Why must magnetic field lines always form closed loops, while electric field lines can begin and end on charges?

P18.3

An electron and a proton move at the same speed perpendicular to the same uniform field. Compare the radii of their circles, their periods, and their senses of rotation.

P18.4

(a) Can a stationary charge feel a magnetic force? (b) Can a moving charge pass through a magnetic field without being deflected? Explain each.

LEVEL B – STANDARD PROBLEMS

P18.5

An alpha particle (charge $+2e$, mass 6.64×10^{-27} kg) moves at 1.0×10^6 m/s perpendicular to a 0.20 T field. Find the radius and period of its circular motion.

P18.6

An electron moving at 5.0×10^6 m/s enters a 0.010 T field with its velocity at 30° to the field. Find the radius and the pitch of its helical path.

P18.7

A straight wire 0.50 m long carries 3.0 A at 40° to a uniform 0.80 T field. Find the magnitude of the force on it, and describe its direction.

P18.8

A 50 g metal rod, 0.20 m long, slides on frictionless horizontal rails 0.20 m apart. A vertical magnetic field of 0.50 T is present, and a current of 4.0 A flows through the rod. Find the rod's acceleration. How far does it travel from rest in 0.50 s?

P18.9

A square coil of 50 turns, 10 cm on a side, carries 0.50 A in a 0.30 T field. The normal to the coil makes 30° with the field. Find the torque on the coil, and the work needed to rotate it from the aligned position to this angle.

P18.10

Uranium enrichment by mass spectrometry. Singly charged ^{235}U and ^{238}U ions are accelerated through 2.0 kV and bent in a 0.25 T field. Find the radius for each, and their separation after a semicircle. ($1\text{ u} = 1.66 \times 10^{-27}\text{ kg}$.)

P18.11

Electromagnetic blood flowmeter. An artery of diameter 5.0 mm lies in a 0.040 T field, and electrodes on opposite sides of the vessel measure 80 μV . Find the blood speed and the volume flow rate in mL/min.

P18.12

MRI physics. (a) Find the proton Larmor frequency at 1.5 T and at 3.0 T. (b) A gradient coil adds a field that changes by 10 mT per metre along the patient's body. By how much does the resonant frequency differ between two slices of tissue 1.0 mm apart? (c) Explain how this allows the scanner to tell where a signal comes from.

LEVEL C – CHALLENGE PROBLEMS

P18.13

The cyclotron. (a) Show that the orbital period of a non-relativistic particle in a uniform field does not depend on its speed, and explain why this makes the cyclotron possible. (b) Derive $K_{\max} = q^2 B^2 R^2 / 2m$. (c) A deuteron (charge e , mass 3.34×10^{-27} kg) is accelerated in a cyclotron with $B = 1.5$ T and $R = 0.60$ m. Find the oscillator frequency and the final energy. (d) Why does a simple cyclotron fail at very high energies, and what is done about it?

P18.14

Hall effect. (a) Derive $V_H = IB/(nqt)$ for a strip of thickness t (measured along $\vec{B}$). (b) Find V_H for a copper strip 0.10 mm thick ($n = 8.5 \times 10^{28} \text{ m}^{-3}$) carrying 10 A in a 1.0 T field. (c) Repeat for a doped silicon strip with $n = 1.0 \times 10^{21} \text{ m}^{-3}$, the same thickness, and 1.0 mA of current. Why are Hall sensors made from semiconductors?

P18.15

A wire carrying current I is bent into a semicircle of radius R and lies in a plane perpendicular to a uniform field B . (a) By integrating $I d\vec{l} \times \vec{B}$, show that the net force on it has magnitude $2IRB$, the same as for a straight wire joining its ends. (b) Use this idea to explain why the net force on any closed current loop in a uniform field is zero.

P18.16

Spin and MRI signal. A proton's magnetic moment is 1.41×10^{-26} J/T. (a) Find the energy difference between a proton aligned with and against a 3.0 T field. Show that $\Delta E/\hbar$ equals the Larmor frequency. ($\hbar = 6.63 \times 10^{-34}$ J s.) (b) At body temperature, the fractional excess of protons aligned with the field is about $\mu B/kT$. Evaluate it, and explain why MRI signals are weak and why stronger magnets give better images.

P18.17

Crossed fields. A proton starts from rest at the origin in a region with a uniform electric field $\vec{E} = E\hat{j}$ and a uniform magnetic field $\vec{B} = B\hat{k}$. (a) Explain qualitatively why the proton does not simply accelerate along $\vec{E}$. (b) Show that its average motion is a drift along $+\hat{x}$ at speed E/B , and that its maximum speed is $2E/B$. (c) Evaluate the drift speed for $E = 1.0 \times 10^4$ V/m and $B = 0.10$ T.

P18.18

The radiation belts. Protons with a kinetic energy of 1.0 MeV are trapped in the Earth's field at about three Earth radii from the centre, where $B \approx 1.1 \times 10^{-6}$ T. (a) Find the radius of the circle each proton makes around a field line. (b) Find its cyclotron period. (c) Explain qualitatively why such particles bounce back and forth between the north and south polar regions, and why the aurora appears in rings around the magnetic poles.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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